

File Permission

A Comprehensive Exercise Set for Managing Permissions and Ownership

Linux File Permissions Exercise Set

This exercise set will guide you through managing file permissions, ownership, and special permissions on Linux systems. Each section includes tasks to reinforce your understanding.

1. Checking Permissions

Instructions

1. **Create a file and a directory:**
2. `touch myfile.txt`
`mkdir mydir`
3. **Check the permissions of the file and directory:**
4. `ls -l myfile.txt mydir`

Task

- Identify the owner, group, and permissions from the output of `ls -l`. The output format will show:
 - File type and permissions (e.g., `-rw-r--r--`)
 - Number of links
 - Owner name
 - Group name
 - File size
 - Last modified date and time
 - File or directory name

2. Changing Permissions (Symbolic Mode)

Instructions

- **Add execute permission for the user on myfile.txt:**
- `chmod u+x myfile.txt`

- **Remove write permission for the group on myfile.txt:**
- `chmod g-w myfile.txt`
- **Set read-only permission for others on myfile.txt:**
- `chmod o=r myfile.txt`
- **Set full access for user, read/execute for group, and no access for others on myfile.txt:**
- `chmod u=rwx,g=rx,o= myfile.txt`

Task

- Verify your changes using `ls -l`.

3. Changing Permissions (Numeric Mode)

Instructions

- **Set permissions 755 on mydir (full for user, read/execute for group & others):**
- `chmod 755 mydir`
- **Set permissions 644 on myfile.txt (read/write for user, read-only for group & others):**
- `chmod 644 myfile.txt`
- **Set permissions 700 on myfile.txt (full for user, no access for others):**
- `chmod 700 myfile.txt`

Task

- Check and explain what changed in the file permissions.

4. Changing Ownership

Instructions

1. **Create a new user and group:**
2. `sudo useradd testuser`
`sudo groupadd testgroup`
3. **Change the owner of myfile.txt to testuser:**
4. `sudo chown testuser myfile.txt`
5. **Change the group of myfile.txt to testgroup:**
6. `sudo chown :testgroup myfile.txt`
7. **Change both owner and group of mydir to testuser:testgroup:**
8. `sudo chown testuser:testgroup mydir`

Task

- Use `ls -l` to verify ownership changes.

5. Changing Group Ownership

Instructions

- **Change the group of myfile.txt back to your default group:**
- `sudo chgrp yourgroup myfile.txt`
- **Change the group of mydir recursively:**
- `sudo chgrp -R yourgroup mydir`

Task

- Verify with `ls -l`.

6. Special Permissions

Instructions

1. **Set SetUID on an executable file:**
2. `echo -e "#!/bin/bash\nnecho Hello" > script.sh`
`chmod u+s script.sh`
3. **Set SetGID on a directory:**
4. `chmod g+s mydir`
5. **Set Sticky bit on a directory:**
6. `chmod +t mydir`

Task

- Explain how these special permissions affect files inside the directory. SetUID allows a program to run with the permissions of the file owner. SetGID ensures new files in the directory inherit the directory's group. The Sticky bit prevents users from deleting files they do not own in a shared directory.

7. Default Permissions (umask)

Instructions

1. **Check current umask:**
2. `umask`
3. **Create a new file and directory. Check permissions:**
4. `touch newfile.txt`
`mkdir newdir`
`ls -l`
5. **Set a new umask 027 and create another file/directory. Compare:**
6. `umask 027`
`touch anotherfile.txt`
`mkdir anotherdir`
`ls -l`

Task

- Compare the default permissions of files and directories created with different umasks. The umask dictates the default permission set for newly created files and directories by

subtracting its value from the system's default.